Cross-validation under data dependence: a review-derived taxonomy and trustcv, a leakage-aware Python toolkit

Abdolamir Karbalaie
abdolamir.karbalaie@ki.se

Karolinska Institutet
Farhad Abtahi
Karolinska Institutet

Article

Keywords: Data leakage, Grouped data, Temporal validation, Spatial validation, Machine learning

Posted Date: April 15th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-9357577/v1

Additional Declarations: No competing interests reported.

Cross-validation under data dependence: a review-derived taxonomy and trustcv, a leakage-aware Python toolkit

Abdolamir Karbalaie[1,*] and Farhad Abtahi[1,2,3]

[1]Department of Clinical Science, Intervention and Technology, Karolinska Institutet, Stockholm 17177, Sweden
[2]Department of Biomedical Engineering and Health Systems, KTH Royal Institute of Technology, Huddinge 14157, Sweden
[3]Department of Clinical Physiology, Karolinska University Hospital, Stockholm 17176, Sweden
[*]Corresponding author: abdolamir.karbalaie@ki.se, farhad.abtahi@ki.se

April 9, 2026

Abstract

Inappropriate cross-validation can inflate performance estimates in medical machine learning, yet validation methods for grouped, temporal, and spatial data remain fragmented across literatures and incompletely supported in existing tools. We addressed this gap through a methodological study combining evidence synthesis, controlled synthetic illustration, and open-source implementation. Using purposive bidirectional snowballing from canonical seed papers, we identified and screened 29 cross-validation methods relevant to medical machine learning and organized them into a four-category taxonomy: independent and identically distributed (i.i.d., n = 9), grouped (n = 8), temporal (n = 8), and spatial (n = 4). Inclusion required documented medical or health-related application, or clear transferability to medical data structures. All 29 methods were implemented in `trustcv` (`https://ki-smile.github.io/trustcv`), a framework-agnostic Python toolkit that provides automated detection of six leakage types and structure-aware validation workflows. Controlled synthetic grouped benchmarks showed that, in the medium-leakage scenario, ignoring patient-level structure inflated area under the receiver operating characteristic curve (AUC) relative to Group K-Fold by 18.2 and 11.7 percentage points at the observation and patient levels, respectively. Implementation reliability was confirmed through automated verification of key correctness properties across 141 unit tests in 11 test modules. Together, the taxonomy and toolkit provide a practical foundation for more reliable, structure-aware model evaluation in medical machine learning.
keywords: Data leakage; Grouped data; Temporal validation; Spatial validation; Machine learning

Introduction

Machine learning is now routinely applied to clinical datasets for disease diagnosis [1, 2, 3], risk stratification [4], and treatment optimisation [5, 6], with numerous clinical applications demonstrating both the promise and risks of data-driven decision support. However, recent reviews show that many published medical ML models are not ready for clinical translation, with poor evaluation design and data leakage among the recurrent causes [7, 8]. Kapoor and Narayanan [9] identified data leakage in 294 papers across 17 scientific fields and described eight distinct leakage mechanisms, showing that methodological errors are widespread across ML-based research. In the medical domain specifically, Roberts et al. [7] screened 2,212 studies on ML models for COVID-19 diagnosis and prognosis, selected 62 for detailed quality assessment, and found that none were of potential clinical use because of methodological flaws and underlying bias. Common problems included lack of external validation, inappropriate handling of multiple images per patient, and failure to account for temporal structure in the data. Similarly, Varoquaux and Cheplygina [8] documented pervasive methodological weaknesses in medical imaging ML, showing that validation strategies associated with inflated performance estimates are disproportionately represented in the literature.

Data leakage represents a common and consequential problem in medical ML. Review and case-study literature has documented leakage problems in medical ML, including patient-level leakage (i.e., the same patient appearing in both training and test sets) and preprocessing leakage (i.e., fitting transformers on the full dataset before splitting) [8, 10]. Preprocessing leakage is also a documented problem in medical ML, for example, applying over-sampling before train/test partitioning can produce overly optimistic results [11]. These errors can substantially inflate performance metrics, creating models that appear clinically useful but fail upon deployment.

Saeb et al.[10] provided a compelling demonstration using the UCI Human Activity Recognition dataset with 30 subjects. Record-wise cross-validation (i.e., splitting individual observations irrespective of subject identity) yielded classification error rates of approximately 2%, whereas subject-wise cross-validation (i.e., ensuring all observations from a given subject appear in the same fold) yielded error rates of approximately 7% with 30 subjects and up to 27% with only 2 subjects. These results demonstrate that record-wise evaluation substantially overestimates model performance by exploiting patient-specific patterns rather than learning generalisable activity signatures. Kaufman et al. [12] formalised leakage in data mining as the introduction of information not legitimately available at prediction time and described how it can arise through data handling and preprocessing choices.

Medical datasets frequently violate the independent and identically distributed (i.i.d.) assumption that underlies standard cross-validation methods. Three primary sources of dependency create challenges.

Patient-level dependencies. Medical datasets typically contain multiple observations per patient, particularly in longitudinal and dynamic prediction settings [13]. When repeated observations from the same patient appear in both training and test sets, models can exploit patient-specific patterns and produce inflated performance estimates [10].

Temporal structure. Clinical data are inherently temporal, with observations ordered by time of collection. Models trained on later data and tested on earlier observations commit temporal leakage by using future information unavailable at the time of prediction. This is particularly problematic for forecasting applications such as disease progression prediction or early warning systems [14].

Spatial autocorrelation. Geographic health data in epidemiology and environmental health shows spatial dependencies where nearby locations have similar characteristics [15, 16]. Roberts et al.[15] demonstrated in ecological species distribution modelling that standard cross-validation substantially overestimates model performance compared to spatially-blocked approaches, a finding with direct implications for spatial epidemiological applications. Ploton et al.[16] quantified this effect, showing that random cross-validation yielded an R-squared of 0.53, which dropped to approximately 0.15 under spatial cross-validation with appropriate buffer distances.

These methodological failures occur against a backdrop of clear regulatory and reporting expectations. The FDA's Good Machine Learning Practice (GMLP) principles [17], developed with Health Canada and the UK's MHRA, include Principle 4 ('Training Data Sets Are Independent of Test Sets') and Principle 8 on statistically sound testing under clinically relevant conditions. The European CE Medical Device Regulation (MDR) 2017/745 [18] requires comprehensive technical documentation including clinical evaluation reports. The TRIPOD+AI statement [19] provides reporting guidance for clinical prediction models developed with regression or machine learning methods, including documentation of internal validation procedures and model performance estimates. Together, these frameworks create an expectation that validation design be structure-aware and transparently documented — yet, as the evidence above shows, inappropriate validation and undocumented leakage remain common in practice. A contributing factor is that the tools available to practitioners do not provide unified, structure-aware validation support with automated leakage detection.

scikit-learn [20] provides the methodological foundation for ML validation in Python, offering 15 cross-validation splitters as of version 1.6: basic methods (KFold, StratifiedKFold, ShuffleSplit, StratifiedShuffleSplit, RepeatedKFold, RepeatedStratifiedKFold), leave-one-out methods (LeaveOneOut, LeavePOut), grouped methods (Group K-Fold, Stratified Group K-Fold, Leave One Group Out (LOGO), Leave P Groups Out, Group Shuffle Split), time series methods (TimeSeriesSplit), and PredefinedSplit. However, scikit-learn lacks spatial validation methods, purged temporal validation with embargo periods, bootstrap validation as a splitter class, hierarchical grouped validation, and any form of automatic leakage detection. MLme [21] provides a user-friendly, classification-focused interface for data exploration, automated model comparison, customizable pipelines, visualization, and standard k-fold evaluation, but it is not presented as a general framework for structured validation in grouped, temporal, or spatial data settings. In the R ecosystem, tidymodels [22] provides a consistent framework for modeling and machine learning using tidyverse principles, and mlr3 [23, 24] provides a modern framework for resampling and nested resampling, while specialized spatiotemporal resampling is available through mlr3spatiotempcv. However, these tools require R proficiency, lack Python integration for deep learning workflows common in medical imaging, and do not provide automatic leakage detection. mlr3spatiotempcv provides robust spatiotemporal resampling for geospatial applications but does not natively address patient-level dependencies common in clinical data and provides visual rather than algorithmic verification of independence, which is insufficient for regulatory-grade medical software.

Methods relevant to grouped, temporal, and spatial validation remain fragmented across multiple literatures and have not been synthesised for medical ML practitioners. To address this, we conducted a purposive snowball sampling review of the medical ML validation literature, tracing how cross-validation methods have been developed, adapted, and applied across fields. This study is therefore best understood as a methodological study with two linked outputs: a structured evidence synthesis and an open-source implementation that makes the synthesis operationally usable. Rather than reporting findings as a list, we implemented all identified methods in a unified open-source toolkit, making the taxonomy operationally usable for practitioners.

This work makes two primary contributions. The first is a purposive snowball sampling review identifying 29 cross-validation methods relevant to medical ML, organised into a four-category taxonomy by data structure (i.i.d., grouped, temporal, spatial). To the authors' knowledge, this is the first such consolidated taxonomy for the medical ML context. The second is `trustcv`, a framework-agnostic Python toolkit that translates the taxonomy into a unified, tested implementation. The toolkit provides automated data leakage detection enabled by default, comprehensive educational materials, decision guides, interactive notebooks, and annotated clinical scenario scripts. It additionally generates structured validation documentation with content mapped to FDA GMLP and TRIPOD+AI requirements, providing a starting point for the validation methodology component of regulatory technical files; such outputs should be interpreted as methodological documentation support rather than substitutes for full regulatory evidence packages [19]. Rather than introducing new splitting algorithms, trustcv's contribution lies in unifying established methods under a leakage-aware, structure-driven interface.

Accordingly, this study had four aims: (1) to identify cross-validation strategies relevant to

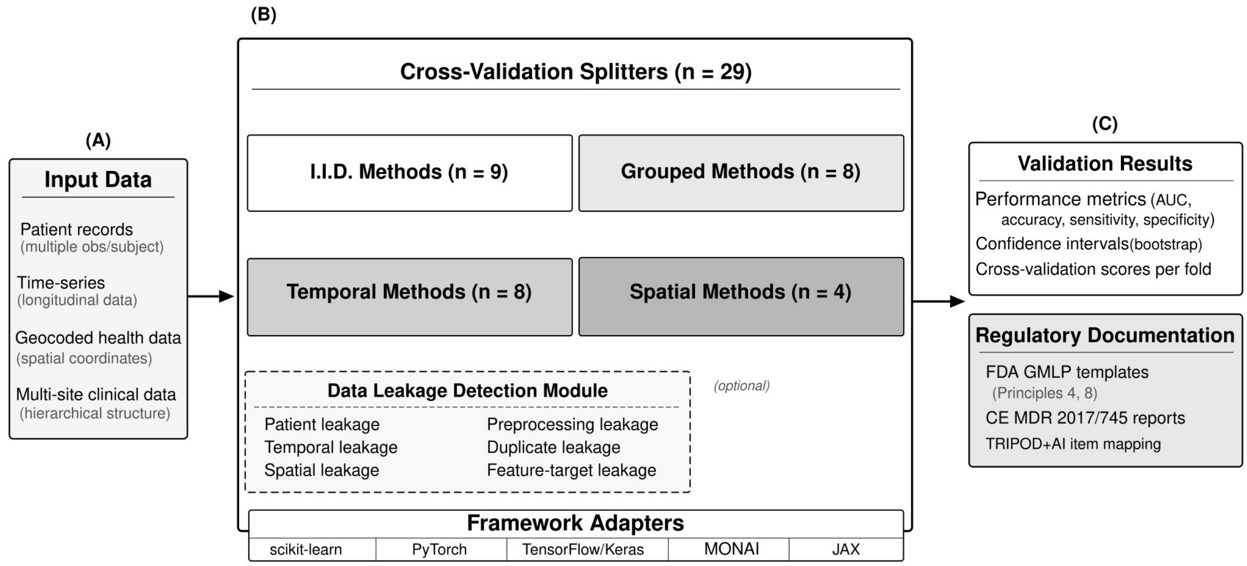

Figure 1: Schematic overview of `trustcv` architecture and cross-validation method taxonomy. (A) Medical data structures motivating the four-category taxonomy: patient-level grouping, temporal ordering, and spatial autocorrelation each require dedicated cross-validation approaches. (B) The 29 methods identified through the snowball sampling review, organised into four categories based on data structural properties; full method descriptions and references are provided in Tables 1–4. (C) Intended toolkit outputs including performance summaries and documentation templates with content mapped to regulatory frameworks. Input data flows through the Splitters module; data leakage detection is enabled by default. This figure summarises toolkit design and intended workflow; comparative benchmarking results are presented in Table 5.

medical ML through purposive bidirectional snowballing from canonical seed papers, (2) to organise the retained methods into a data-structure-driven taxonomy, (3) to translate that taxonomy into a unified Python toolkit with leakage-aware defaults, and (4) to illustrate, in controlled synthetic structured-data settings, how validation-strategy choice affects estimated model performance.

Results

Review yield and four-category taxonomy

The structured snowballing process identified 29 cross-validation methods relevant to medical machine learning after harmonisation of naming variants and conceptually overlapping labels. The retained methods were organised into a four-category taxonomy according to the primary data dependency they address: i.i.d. methods (n = 9), grouped methods (n = 8), temporal methods (n = 8), and spatial methods (n = 4). This taxonomy provides a consolidated view of validation strategies relevant to medical machine learning and serves as the conceptual basis for the toolkit implementation (Figures 1–2; Tables 1–4).

Rather than treating method choice as a generic model-evaluation step, the taxonomy makes explicit that cross-validation strategy should be selected according to data structure. Grouped methods preserve patient- or site-level independence, temporal methods preserve chronological order and can incorporate purging or embargo periods, and spatial methods address geographic autocorrelation. Detailed method descriptions, representative medical applications, and key references are provided in Tables 1–4.

The review also revealed substantial terminological fragmentation in the literature. Equivalent or near-equivalent procedures were frequently described under different names depending on disciplinary origin, including multiple aliases for grouped, temporal, and spatial validation strategies. Harmonisation of these naming variants into standardized method labels enabled a consistent taxonomy and direct mapping to software implementations (Supplementary Table S7). This normalization also made comparison with existing tools more interpretable, showing that `trustcv` provides broader coverage across grouped, temporal, and spatial methods than the evaluated alternatives, while uniquely combining this breadth with automated leakage detection and documentation-oriented outputs (Supplementary Table S2). The review also revealed a notable pattern in coverage: the majority of temporal and spatial methods had no identified medical application within the review period (Tables 3–4).

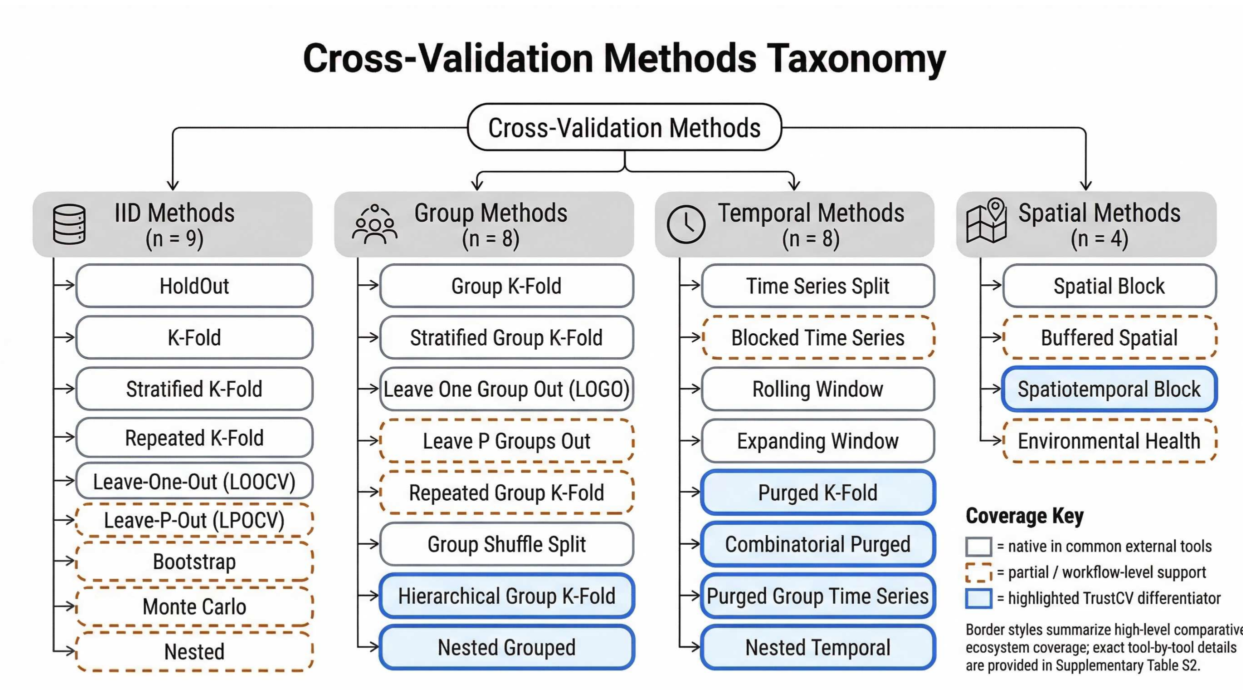

Figure 2: Taxonomy of cross-validation methods for machine learning model evaluation, organised by the primary data dependency each method addresses. The four categories span independent and identically distributed (i.i.d.) methods ($n = 9$, assuming sample independence), grouped methods ($n = 8$, preserving group integrity), temporal methods ($n = 8$, respecting time ordering), and spatial methods ($n = 4$, addressing geographic autocorrelation). Border styles encode comparative ecosystem coverage: solid borders indicate methods natively supported in common external tools (e.g., scikit-learn); dashed borders indicate partial or workflow-level support; solid blue borders highlight methods that represent distinctive contributions of `trustcv` relative to evaluated alternatives. Exact tool-by-tool coverage details are provided in Supplementary Table S2.

Toolkit translation of the review findings

All retained methods were implemented in `trustcv` through a unified Python interface spanning i.i.d., grouped, temporal, and spatial validation settings (Figure 1). Supplementary Tables S1 and S4–S6 summarise parameters, interactive notebooks, example scripts, and framework-specific adapter support across scikit-learn, PyTorch, TensorFlow/Keras, MONAI, and JAX. The decision flowchart in Figure 3 translates the taxonomy into a structured method-selection guide for practi-

Table 1: independent and identically distributed (IID). cross-validation methods implemented in `trustcv`. Methods and references were identified through the snowball sampling review.

Method	Description	Medical Applications	Key References
Hold-Out	Partitions data into separate training and test sets using a single split; the optimal training proportion depends on sample size and signal strength, although split-sample designs commonly allocate most observations to training. Typically uses 70–80% for training.	Toxicity prediction from high-throughput screening assays [25]; large screening mammography datasets [26]; high-dimensional gene-expression studies [27]	Dobbin & Simon [27]; Hastie et al. [28]
K-Fold	Divides the dataset into k equal-sized folds, using each fold once as a test set while training on the remaining k-1 folds. Standard choice is k=5 or k=10, balancing bias-variance tradeoff.	Pathologic diagnosis using whole-slide images[29]; heart disease prediction[30]; diabetes diagnosis[31]	Kohavi[32]; Stone[33]
Stratified K-Fold	Extension of K-Fold that preserves the percentage of samples for each class in every fold. Essential when class distributions are imbalanced.	Cancer subtype classification [34]; adverse drug reaction prediction [35]	Kohavi [32]; Forman & Scholz [36]
Repeated K-Fold	Performs KFold multiple times (typically 5–10 repetitions) with different random seeds. Reduces variance by averaging across randomisations.	Biomarker validation[37]; cardiovascular risk prediction[38]	Molinaro et al. [39]; Kim [40]
LOOCV	Leave-one-out uses n-1 samples for training and 1 for testing, repeated n times. Nearly unbiased but high variance.	biomarker patients with low-grade dysplasia[41]	Cawley & Talbot [42]
LPOCV	Leave-p-out evaluates all C(n,p) combinations. Computationally expensive but exhaustive.	predicts chemotherapy benefit in gastric cancer[43]; evaluating diagnostic systems[44]	Celisse & Robin [45]; Shao [46]
Bootstrap-Validation	Creates training sets by sampling with replacement, using out-of-bag samples for testing. The .632+ correction addresses optimistic bias.	Clinical prediction model development and internal validation[47]; microarray in genomics [48]; ecological and environmental modeling [49]	Efron & Tibshirani[50]; Efron[51]
MonteCarlo	Randomly splits data multiple times (50–200 iterations), allowing flexible split proportions. Also known as shuffle-split validation.	sensitivity analysis in medical image segmentation[52]; feature-selection bias in microarray classification [53]; large-scale patient complaint triage [54]	Xu & Liang [55]; Arlot & Celisse [56]
Nested	Two-loop CV where outer loop evaluates generalisation and inner loop performs hyperparameter tuning. Prevents optimistic bias from using same data for both.	Radiomics and precision oncology[57]; radiomics [58]; Chemometric and metabolomic pipelines [59]	Varma & Simon [60]; Cawley & Talbot [42]

Acronyms: CV = cross-validation.

tioners.

In addition to method execution and leakage checks, `trustcv` generates structured validation summaries and templated reporting outputs intended to support transparent documentation of validation design, split integrity, and model performance. The current implementation includes output mappings relevant to FDA GMLP, CE MDR 2017/745, and TRIPOD+AI-style reporting, with the detailed correspondence summarized in Supplementary Table S3.

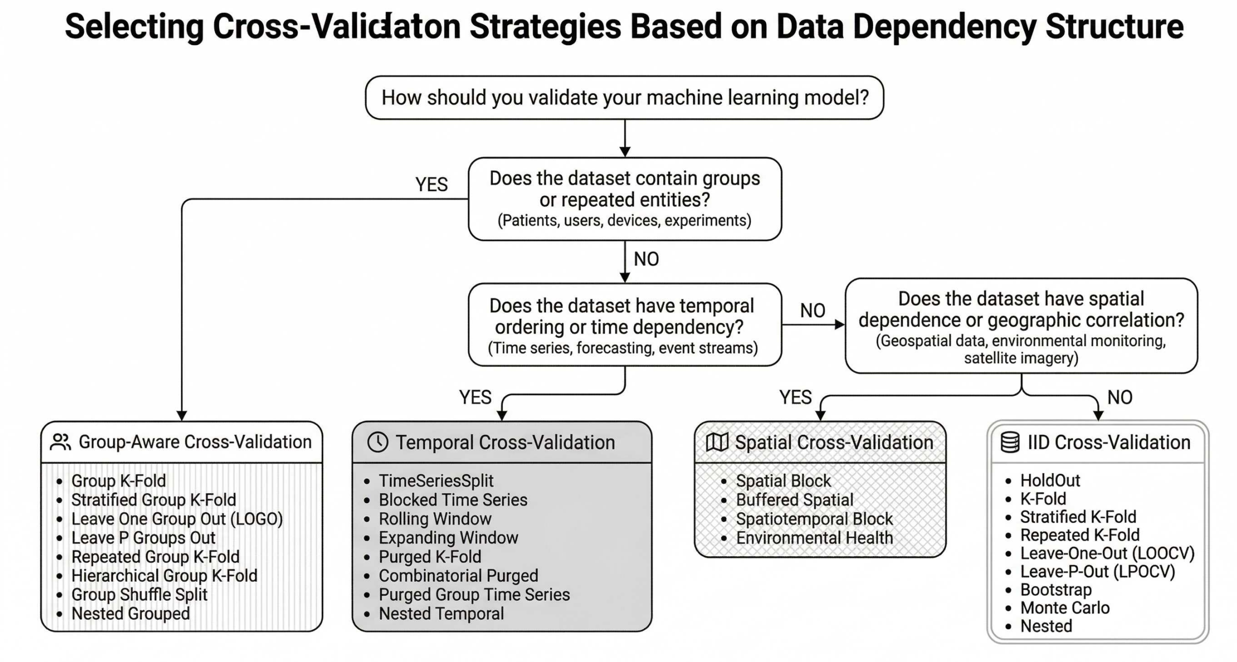

Figure 3: Decision flowchart for selecting cross-validation strategies based on data structure. The flowchart guides practitioners through three sequential questions: (1) whether the dataset contains repeated measurements or group identifiers (e.g., patients, subjects, devices), leading to group-aware methods; (2) whether temporal ordering or time dependence must be respected, leading to temporal methods; and (3) whether the dataset exhibits spatial autocorrelation, leading to spatial methods. If none of these structural dependencies apply, standard i.i.d. methods are appropriate. Advanced evaluation strategies such as nested cross-validation, external validation, and cross-study validation can be applied in combination with any category. A detailed version with method-level recommendations is provided in Supplementary Figure S1.

Automated detection of data leakage

`trustcv` implements automated detection of six leakage categories that are particularly relevant to medical machine-learning pipelines: patient leakage, temporal leakage, spatial leakage, preprocessing leakage, duplicate leakage, and feature-target leakage (Supplementary Table S11). Leakage detection is enabled by default and can be disabled only by explicitly setting `check_leakage=False`, shifting validation safety from an opt-in to an opt-out model.

When enabled, the checker issues warnings for potential leakage and raises errors for confirmed violations under rule-based checks such as patient-identifier overlap or temporal look-ahead. This safety layer operationalizes one of the paper's central practical messages: inappropriate validation often arises from structural violations in how data are partitioned and processed, rather than from the choice of estimator alone.

Controlled synthetic grouped benchmark

Table 5 highlights a clear difference between K-Fold and the group-aware methods in the medium-leakage scenario. At the observation level, K-Fold yielded the highest AUC (0.840), whereas Group K-Fold, Stratified Group K-Fold, LOGO, and Nested Grouped CV were closely grouped between 0.653 and 0.672. The same pattern was observed at the patient level, where K-Fold reached 0.852 and the grouped methods ranged from 0.725 to 0.746. The gap relative to Group K-Fold was larger for K-Fold (+18.2 percentage points (pp) at the observation level; +11.7 pp at the patient level) than for any group-aware method, whose AUC differences remained close to zero. Patient-level estimates were consistently higher than the corresponding observation-level estimates, but the ordering across methods was unchanged. The leakage checker flagged only K-Fold. The inflation magnitude scaled with leakage severity: under high-leakage conditions the K-Fold/Group K-Fold gap increased to 31.3 pp at the observation level, whereas under low-leakage conditions it decreased to 9.7 pp (Supplementary Table S10). Full results for all leakage scenarios are provided in Supplementary Table S10.

Software verification and implementation reliability

Implementation reliability was evaluated using an automated unit-test suite covering splitters, leakage detection, and core workflow integrations. Verified properties included partition completeness, train–test disjointness, group exclusivity for grouped methods, and temporal ordering for time-aware splitters (Supplementary Table S12).

Controlled evaluations of the `DataLeakageChecker` showed correct detection of patient, temporal, and duplicate leakage in the characterised test scenarios, which used clear, clean injection of leakage signatures, no false positives were observed in the corresponding clean settings under these conditions. Together, these results support the claim that the review-derived taxonomy has been translated into a functioning and internally consistent software implementation. These characterizations should be interpreted as targeted correctness checks rather than exhaustive sensitivity analyses; real-world leakage may involve subtler or mixed-mode contamination not represented in the present test scenarios.

Table 2: Grouped cross-validation methods implemented in `trustcv`. Methods and references were identified through the snowball sampling review.

Method	**Description**	**Medical Applications**	**Key References**
Group K-Fold	Extends KFold to ensure all samples from the same group (e.g., patient) appear in the same fold, providing patient-wise k-fold evaluation and preventing data leakage when multiple correlated samples exist per patient.	Multi-slice imaging [61]; Dementia Multi-omics Stratification [62]; Amyloid Plaque Segmentation [63]; BCI EEG Classification [64];	Pedregosa et al. [20]; Saeb et a l.[10]
Stratified Group K-Fold	Combines group constraints with stratification, preserving class distribution across folds while keeping all samples from each group in a single fold.	hEDS Genetic Risk Modeling [65]; ACLR Fear-of-Reinjury [66]; Liver Failure Coagulation Risk [67]; AD Progression Prediction [68]; Patient-level imbalanced CKD classification [69]; Breast lesion classification [70]; Environmental monitoring [71]	Pedregosa et al. [20]
Leave One Group Out (LOGO)	Uses each group (e.g., patient, site, or scanner) as the test set exactly once, training on all remaining groups. Simulates generalising to completely new patients or sites.	External validation of prognostic models [72]; EEG Schizophrenia Diagnosis [73]; Microsleep EEG Classification [74]; Epilepsy Zone Localization [75]; New patient generalisation [76]; Cross-hospital generalisation assessment [77]; Movement patterns fear of re-injury [78, 66]	Saeb et al.[10]; Little et al. [79]
Leave P Groups Out	Leaves p entire groups (e.g., hospitals or regions) out simultaneously for testing, training on the remaining groups. Evaluates generalisation to multiple unseen centres or regions.	Multi-institutional federated learning settings [80]	Riley et al. [81]; Debray et al. [82]
Repeated Group K-Fold	Performs Group K-Fold multiple times with different random group-to-fold assignments (conceptually combining Group K-Fold and repeated resampling). Reduces variance of performance estimates while maintaining group integrity.	Multimodal longitudinal disease progression modelling [83]	Pedregosa et al. [20]; Molinaro et al. [39].
Hierarchical Group K-Fold	Handles nested group structures (e.g., Hospital → Department → Patient) by respecting hierarchy during fold assignment.	Multicenter ward deterioration prediction [84]; Precision oncology basket and umbrella trials [85]; Healthcare predictive analytics implementation [86]	Goldstein et al. [38]
Group Shuffle Split	Randomly partitions groups into training and test sets, preserving group integrity while allowing flexible train/test proportions. Useful for patient-wise random splits and cohort-level resampling without fixed folds.	No medical applications identified within the review period (2000–early 2026)	Pedregosa et al.[20]
Nested Grouped	Combines nested CV with group constraints in both inner and outer loops. Prevents both tuning leakage and group leakage simultaneously.	ACLR Fear-of-Reinjury [66]; EEG Markers of Schizophrenia [87]	Karbalaie & Abtahi [66]

Acronyms: BCI = brain–computer interface; EEG = electroencephalography; hEDS = hypermobile Ehlers–Danlos syndrome. ACLR = anterior cruciate ligament reconstruction; AD = Alzheimer's disease; CKD = chronic kidney disease; CV = cross-validation.

Table 3: Temporal cross-validation methods implemented in `trustcv`. Methods and references were identified through the snowball sampling review.

Method	**Description**	**Medical Applications**	**Key References**
Time Series Split	Forward-chaining validation where training sets grow incrementally and test sets follow in temporal order. Simulates realistic deployment where only historical data is available.	Emergency Department demand forecasting [88]; Cancer patient acute care [89]	Hyndman & Athanasopoulos [90]; Tashman [91]
Blocked Time Series	Divides time series into non-overlapping temporal blocks, using each as a test set while training on non-adjacent blocks.	EEG Mental Workload Classification [92]; EEG Passive BCI Classification [93]; Psychological Time Series Model Selection [94]	Bergmeir & Benítez[95]; Bergmeir et al. [14]
Rolling Window	Uses fixed-size sliding windows for both training and testing. Assumes model relevance is limited to recent history.	ICU HDF Volume Prediction [96]; Crowding Forecasting [97]; COVID-19 Epidemiological Forecasting [98]; ED Demand Forecasting [88]; Heart Transplant Mortality Prediction [99]; COVID-19 Hospital Admissions Forecasting [100]; Postprandial Glucose Prediction [101]; TB Incidence Surveillance [102];	Cerqueira et al.[103]
Expanding Window	Training window grows from minimum size, expanding to include all available past data. Balances recency and sample size.	ICU Dynamic Mortality Prediction [104];	Tashman[91]
Purged K-Fold	Applies K-Fold-style splitting with temporal gaps (purge periods) between training and test sets. Essential when target labels span time periods.	No medical applications identified within the review period (2000–early 2026)	de Prado [105]
Combinatorial Purged	Generates multiple train/test combinations with purge and embargo periods. Provides comprehensive temporal validation.	No medical applications identified within the review period (2000–early 2026)	de Prado [105]; Bailey et al. [106]
Purged Group Time Series	Combines temporal forward-chaining with group constraints and purge gaps. Most rigorous method for longitudinal patient data.	ICU Mortality Prediction (Nested) [107]	de Prado [105]
Nested Temporal	Nested CV respecting temporal order in both loops. Prevents both temporal leakage and tuning bias.	Wearable Cardiorespiratory Monitoring [108]; ICU LOS and Mortality Prediction [109]	Bergmeir & Benítez [95]

Acronyms: EEG = electroencephalography; BCI = brain–computer interface; ICU = intensive care unit. HDF = hemodiafiltration; ED = emergency department; TB = tuberculosis; LOS = length of stay.

Table 4: Spatial cross-validation methods implemented in `trustcv`. Methods and references were identified through the snowball sampling review.

Method	Description	Medical Applications	Key References
Spatial Block	Divides geographic space into non-overlapping rectangular or hexagonal blocks. Block size should exceed the spatial autocorrelation range.	Disease Ecology Prediction [110]; MRI Quality Control [111]; Heart Failure Risk Prediction [112]; Psychosis Outcome Prediction [113];	Roberts et al.[15]; Valavi et al.[114]
Buffered Spatial	Extends Spatial Block CV by adding buffer zones around test blocks. Samples within buffer distance are excluded from training.	Disease Ecology Prediction [110]	Ploton et al.[16]; Le Rest et al.[115]
Spatiotemporal Block	Combines spatial blocking with temporal blocking for data with both spatial and temporal structure. Essential for dynamic spatial processes.	Global NO2 Exposure Modeling [116]; National Air Pollution Exposure [117];	Meyer et al.[118]; Meyer & Pebesma[119]
Environmental Health	Specialised for environmental exposure studies with consideration for exposure gradients, population density, and confounding spatial variables.	Air pollution [120, 121]	Kumar et al.[122]

Acronyms: MRI = magnetic resonance imaging; NO_2 = nitrogen dioxide; CV = cross-validation.

Table 5: Performance of grouped and non-grouped cross-validation strategies under the medium-leakage synthetic grouped scenario, shown at both the observation level (Panel A; n = 1,847 observations) and the patient level (Panel B; n = 500 patients). Group K-Fold is shown as the patient-aware reference. Standard K-Fold ignores patient grouping and therefore permits patient-level leakage.

CV method	AUC (95% CI)	Sensitivity	Specificity	Youden's J	ΔAUC vs Group K-Fold	Leakage checker
Panel A. Observation-level performance ($n = 1{,}847$)						
K-Fold	0.840 [0.822, 0.861]	0.688	0.851	0.539	+0.182	FLAG
Group K-Fold (reference)	0.658 [0.636, 0.677]	0.445	0.755	0.200	Ref	PASS
Stratified Group K-Fold	0.672 [0.630, 0.708]	0.484	0.734	0.219	+0.014	PASS
Leave One Group Out†	0.653 [0.617, 0.687]	0.518	0.701	0.220	-0.005	PASS
Nested Grouped	0.669 [0.651, 0.685]	0.399	0.811	0.210	+0.011	PASS
Panel B. Patient-level performance ($n = 500$)						
K-Fold	0.852 [0.831, 0.874]	0.774	0.957	0.731	+0.117	FLAG
Group K-Fold (reference)	0.734 [0.707, 0.763]	0.484	0.835	0.319	Ref	PASS
Stratified Group K-Fold	0.746 [0.703, 0.785]	0.498	0.806	0.304	+0.011	PASS
Leave One Group Out†	0.725 [0.682, 0.769]	0.516	0.767	0.283	-0.009	PASS
Nested Grouped	0.742 [0.719, 0.765]	0.385	0.892	0.277	+0.007	PASS

Note 1. Standard deviation values across folds and additional metrics are reported in Supplementary Table S10. Full results across low-, medium-, and high-leakage scenarios, including fold-support diagnostics and additional metrics, are provided in Supplementary Tables S9–S10.
Note 2. Stratified Group K-Fold, Leave One Group Out (LOGO), and Nested Grouped preserve group exclusivity. Values are reported as mean AUC with bootstrap 95% confidence intervals where available; LOGO confidence intervals were estimated from pooled out-of-fold predictions across all held-out-patient splits, together with sensitivity, specificity, and Youden's J for descriptive comparison.
Note 3. ΔAUC vs Group K-Fold denotes the absolute difference in mean AUC relative to the patient-aware Group K-Fold reference within the same evaluation level.
† LOGO is included to show the behaviour of a strict group-exclusive estimator. Because each test fold contains a single group, fold-wise AUC estimation was unstable in this simulation, with many single-class test folds; confidence intervals were therefore estimated from pooled out-of-fold predictions across all held-out-patient splits rather than from fold-wise aggregation.
Acronyms: AUC = area under the receiver operating characteristic curve; CI = confidence interval.

Discussion

This study addressed a practical gap in medical ML by combining three linked components: a structured synthesis of cross-validation strategies relevant to medical data, controlled synthetic illustrations of method-selection consequences, and an open-source implementation that makes these strategies operationally usable. Taken together, these components show that the main challenge is not the absence of appropriate validation methods, but the absence of a coherent pathway linking data structure, method selection, implementation, and leakage-aware evaluation. By consolidating 29 methods into a four-category taxonomy and translating them into `trustcv`, the present work provides that pathway in a form intended for direct use in medical ML workflows.

A central implication of the present findings is that cross-validation strategy should be selected according to the structural properties of the data rather than by convention. This principle is often acknowledged in methodological literature, yet it remains inconsistently reflected in applied medical ML, where i.i.d. validation is still frequently used despite the presence of patient-level dependence, temporal ordering, or spatial autocorrelation [9, 8, 15]. The grouped synthetic benchmark provides a concrete illustration of the practical consequence of this mismatch: when patient structure was ignored, estimated discrimination increased markedly relative to group-aware validation. This result should not be interpreted as a universal inflation constant, but as a controlled demonstration of a broader principle: evaluation design materially affects apparent model performance. A related pattern emerging from the review is that most temporal and spatial methods currently lack a documented medical application, suggesting that disciplinary transfer from financial and geospatial literatures into medical ML remains incomplete. This gap reinforces the practical motivation for a consolidated taxonomy: practitioners cannot adopt methods they are unaware of, and the absence of uptake may reflect a discovery barrier as much as a suitability question.

An important contribution of this work is that it does not stop at method synthesis. Knowing that a validation strategy exists in the literature is not equivalent to being able to apply it correctly in practice. Methods such as purged temporal validation, hierarchical grouped splitting, or spatial blocking are easy to misimplement in ad hoc workflows because their defining constraints must be enforced explicitly. In this context, `trustcv` functions as a translation layer between methodological knowledge and applied workflow. Its value lies not in introducing new cross-validation methods, but in reducing the implementation barriers that often prevent appropriate methods from being applied consistently in practice.

The current results also support the argument that leakage detection should be treated as a default safety mechanism rather than an optional utility. In practice, leakage is often inadvertent: users may not recognize that group overlap, temporal look-ahead, duplicate contamination, or preprocessing mistakes have compromised the evaluation design [9, 12]. The default behaviour in `trustcv` operationalises this principle directly, as the controlled detection results above demonstrate.. The controlled detection experiments reported here show that patient, temporal, and duplicate leakage can be detected in the three characterized leakage scenarios, with no false positives in the corresponding clean settings under the designed injection conditions, supporting the feasibility of this approach. At the same time, these findings should be interpreted as proof of principle rather than exhaustive characterization of detection performance across all leakage types and severity levels.

The synthetic benchmark also highlights a practical consideration when applying LOGO with patient-level groups. In the 500-patient simulation, each test fold contained a single patient contributing only 2–5 observations, so many individual test folds contained observations from one class only, rendering fold-wise AUC estimation undefined or degenerate at the fold level. Performance was therefore estimated from pooled out-of-fold predictions aggregated across all 500 held-out patients, rather than by averaging fold-wise AUC values. This pooled approach yielded stable and interpretable estimates: LOGO produced AUC values of 0.653 at the observation level and 0.725 at the patient level Table 5), closely consistent with the other group-aware methods and passing the leakage checker. The issue is therefore not with LOGO as a method—it enforces strict group exclusivity and its performance estimates were aligned with Group K-Fold—but with the assumption that fold-wise AUC aggregation is always viable. In any dataset where each test fold contains very few observations per group, or where class prevalence is highly variable across groups, fold-wise AUC will be unstable. Practitioners using LOGO in such settings should compute performance from pooled out-of-fold predictions rather than averaging fold-level metrics, and should verify that test folds contain sufficient observations from both classes before relying on fold-wise estimates.

The comparison with existing ecosystems clarifies the practical niche of the present work. General frameworks such as `scikit-learn` provide the foundation for many ML workflows, but they do not offer unified support for the full range of grouped, temporal, and spatial strategies relevant to medical datasets, nor do they provide automated leakage detection. Other ecosystems offer strong support for specific subdomains, especially spatiotemporal resampling, but do not directly target the combination of patient-level dependence, longitudinal leakage, and documentation-oriented validation support that is especially relevant in medical ML. Supplementary Table S2 shows that `trustcv` provides broader coverage across grouped, temporal, and spatial methods than the evaluated alternatives. The contribution of `trustcv` is therefore not simply a broader menu of splitters, but a unified implementation that aligns taxonomy, execution, safety checks, and reporting support within a single workflow.

These findings have implications beyond software availability. For practitioners, the four-category taxonomy provides a structured way to diagnose whether the dataset should be treated as i.i.d., grouped, temporal, or spatial before selecting a validation strategy. For peer reviewers and journal editors, it offers a clearer basis for assessing whether the reported evaluation design matches the data structure described in a manuscript. For software developers and methodological researchers, it provides a tested baseline implementation that can support further benchmarking, extension, and workflow integration. The documentation-oriented reporting layer may also assist in preparing transparent validation records. As noted above, such outputs are intended as methodological starting points and do not substitute for full regulatory evidence packages.

Limitations and future directions

The present study has several scope-defining limitations. First, regarding review methodology: The evidence-synthesis component used purposive bidirectional snowballing and therefore remains shaped by the citation networks of the selected seed papers, the English-language literature captured through those networks, and the review cutoff. The review should therefore be interpreted as structured and transparent but not exhaustive in the sense of a full database-driven systematic review.

Second, regarding taxonomy scope: The taxonomy was intentionally defined at the level of canonical validation strategies rather than all possible implementation-level variants. As a result, repeated, shuffled, nested, or hybrid extensions were not always enumerated as separate methods when they reflected the same underlying split logic, and some workflow variants encountered in practice were represented through their core methodological family rather than listed independently. Inclusion in the taxonomy does not imply identical maturity of medical uptake across all 29 methods. Relatedly, the absence of an identified medical application for some listed methods by the review cutoff should not be interpreted as evidence of irrelevance to medical machine learning, but rather as an indication that no eligible application was captured within the present screening and normalization framework.

Third, regarding quantitative illustration: The quantitative illustrations were designed as controlled methodological demonstrations rather than clinical benchmarks, so the reported inflation magnitude should not be assumed to transfer directly to real datasets. The grouped scenario constitutes the main quantitative benchmark reported in the main text; the ICU monitoring, spatial epidemiological, and multisite clinical-trial scenarios were designed as illustrative workflow examples and are therefore not included in the reported results. The present work includes no real-data external validation. Demonstrating that the taxonomy and toolkit behave appropriately on published clinical datasets with known structure such as MIMIC-IV for temporal grouped data, PhysioNet Challenge datasets for ICU time-series, or spatially annotated epidemiological datasets, remains an important unfinished step before the practical generalisability of the implementation can be assessed. The real-data result from Saeb et al. [10], is more directly informative about real-world inflation magnitude than the present synthetic benchmark precisely because it reflects genuine patient variability rather than parameterised simulation.

Finally, regarding toolkit scope: Although `trustcv` detected several leakage types in the characterized test settings, broader performance characterization across mixed contamination regimes remains pending, and the current implementation is limited to Python-based workflows, with some methods remaining computationally demanding for large datasets.

These limitations define clear next steps. A first priority is external evaluation on publicly available clinical datasets with known grouped, temporal, or spatial structure, such as MIMIC-IV [123], PhysioNet Challenge datasets [124] or the UCI Human Activity Recognition dataset [10], in order to assess how the taxonomy and implementation behave under real application conditions. A second is formal characterization of leakage-checker performance across leakage types and severity levels, including mixed-contamination scenarios. A third is extension from method execution toward method recommendation, for example by using dataset metadata and detected structural properties to suggest candidate validation strategies automatically. Longer-term directions include support for non-Euclidean distance matrices, broader integration with additional analysis ecosystems, and more interactive interfaces that make structure-aware validation easier to adopt in routine research workflows.

Conclusion

This study addressed a structural gap in medical machine learning by combining evidence synthesis, controlled synthetic evaluation, and open-source implementation. The main contribution is a data-structure-driven taxonomy of 29 cross-validation methods relevant to medical datasets with i.i.d., grouped, temporal, and spatial structure, together with a unified Python toolkit that makes these methods operationally usable in practice. The grouped synthetic illustration showed that ignoring patient-level structure can substantially inflate apparent model performance under realistic repeated-measures conditions, demonstrating that validation strategy is a methodological choice with measurable consequences for reported performance.

The practical implication is straightforward: data structure should be diagnosed before a validation strategy is chosen. In this context, `trustcv` is intended to reduce the gap between methodological knowledge and correct implementation by combining structure-aware splitters, default leakage detection, and reproducible validation workflows within a single framework. Taken together, the taxonomy and toolkit provide a practical foundation for more reliable model evaluation in medical ML and a starting point for future benchmarking, validation guidance, and workflow integration.

Methods

Study design and scope

This work comprised two linked methodological components: (i) a structured methodological review designed to identify cross-validation strategies relevant to medical ML, and (ii) an open-source software implementation translating the retained methods into a unified Python toolkit. The review focused specifically on validation strategies applicable to medical datasets that depart from the i.i.d. assumption through patient-level grouping, temporal ordering, spatial autocorrelation, or combinations of these structures. Because the terminology of cross-validation is fragmented across statistical learning, medical informatics, ecology, and quantitative finance, we used a purposive bidirectional snowballing strategy rather than a conventional database-only search. The review process followed the snowballing principles described by Wohlin et al. [125] and used PRISMA 2020 as a reporting scaffold for transparent description of study identification, screening, and inclusion decisions rather than as a claim of a full database-driven systematic review [126].

Seed paper selection and bidirectional snowballing

Seven seed papers were selected a priori as starting points for the review on the basis of citation influence, methodological breadth, and relevance to either foundational cross-validation theory or the practical problem of validation bias in medical machine learning: Stone[33], Kohavi[32], Efron and Tibshirani[50], Varma and Simon[60], Roberts et al.[15], Kapoor and Narayanan[9], and Varoquaux and Cheplygina[8]. Collectively, these seed papers span classical statistical validation theory, model-selection bias, spatially structured validation, and contemporary documentation of leakage and evaluation failures in ML-based science and medical imaging.

Backward snowballing was performed by screening the reference lists of the seed papers to recover original methodological descriptions, foundational theoretical papers, and earlier formulations of cross-validation procedures. Forward snowballing was then performed by screening citing papers identified through Google Scholar, Semantic Scholar, and arXiv in order to identify later variants, extensions, and domain-specific adaptations. Forward citation screening covered publications from 2000 through early 2026, whereas older foundational work was primarily recovered through backward snowballing. Snowballing proceeded iteratively until saturation, defined as a complete forward-plus-backward iteration yielding no additional eligible method concepts.

Screening, normalization, and eligibility assessment

Candidate methods identified through snowballing were screened independently by both authors against four predefined eligibility criteria: (1) documented use in medical, clinical, or health-related research, or clear transferability to health-related data structures, (2) explicit relevance to a data dependency commonly encountered in medical ML, including grouped, temporal, spatial, or standard i.i.d. settings, (3) implementability as a reusable resampling or splitting procedure within a general validation interface, and (4) conceptual distinctness from methods already retained in the developing taxonomy.The snowballing process identified a total of 37 candidate method concepts prior to eligibility assessment; of these, 29 were retained after applying the four criteria. Initial independent screening produced agreement on 27 of the 29 retained methods (Cohen's $\kappa = 0.91$, computed across all 37 screened candidate), disagreements on the remaining two were resolved through consensus discussion."

To reduce ambiguity arising from inconsistent terminology, screening was performed at the *method-concept* level rather than the literal-title level. A predefined normalization table mapped each retained method to a standardized search label and a set of naming variants used during literature retrieval and screening (Supplementary Table S7). This step was particularly important for grouped, temporal, and spatial methods, where equivalent or near-equivalent procedures are often described under different domain-specific names. Candidate records referring to the same splitting logic were merged under a single canonical label, whereas procedures were retained as distinct methods only when they imposed meaningfully different partition constraints, leakage barriers, or resampling behaviour.

Data extraction and taxonomy construction

For each retained method, we extracted the following fields: canonical method name; standardized search label; naming variants and aliases; primary data dependency addressed; operational splitting rule; leakage mechanism targeted; typical medical or health-related use case; representative methodological references; representative application references; computational or practical limitations; and implementation feasibility within a common Python API. These extraction fields were used both to construct the narrative taxonomy and to guide software implementation.

Retained methods were assigned to one primary category according to the dominant structural dependency they address: i.i.d., grouped, temporal, or spatial. Methods that primarily serve as *evaluation overlays*, most notably nested validation, were classified within the category in which their operational constraints are enforced. Thus, standard nested cross-validation was retained within the i.i.d. family, whereas *Nested Grouped CV* and *Nested Temporal CV* were retained within the grouped and temporal families, respectively. This decision preserved conceptual clarity while remaining faithful to implementation requirements. The final taxonomy comprised 29 distinct methods across four categories.

Software design goals and implementation strategy

The retained methods were translated into `trustcv`, an open-source Python toolkit designed to make the review-derived taxonomy operationally usable in day-to-day ML workflows. The software design was guided by four goals: broad method coverage across the identified taxonomy, leakage-aware defaults, framework-agnostic usability, and reproducible, documentation-oriented output.

`trustcv` was developed in Python (version 3.9+) and organised around a modular architecture consisting of splitters, leakage checkers, metrics, reporting utilities, and framework-specific integration layers. At the conceptual level, the Splitters module implements the 29 retained cross-validation methods across all four submodules. Splitters expose a scikit-learn-compatible interface based on `get_n_splits()` and `split()`, supporting reuse within standard Python validation pipelines [20]. The Checkers module implements the `DataLeakageChecker` class for automated assessment of common leakage mechanisms. The Metrics module provides task-adaptive performance summaries across classification, regression, and image segmentation workflows; the specific metrics, aggregation strategy, and confidence interval method used in the present experiments are described in the Performance estimation and statistical summary section below. The Reporting module generates structured outputs intended to support transparent validation documentation, including mappings relevant to TRIPOD+AI reporting [19].

Method implementation followed a translation rule designed to preserve the core partition logic described in the literature while exposing a consistent, framework-agnostic API. Where a method was defined in the literature as a conceptual family rather than a standardized software object, we implemented the minimal parameterization necessary to preserve its defining constraints, such as group exclusivity, temporal ordering, purge gaps, embargo intervals, or spatial blocking. `trustcv` class names such as `KFold` and `GroupKFold` intentionally extend familiar scikit-learn naming patterns while allowing harmonized parameters such as `shuffle` and `random_state` where useful for consistency across methods. Core numerical dependencies include NumPy[127] and pandas.[128]

Leakage detection logic

`trustcv` implements automated leakage detection as a default safety layer through the `DataLeakageChecker`, which returns structured warnings, severity-ranked reports, and remediation suggestions. The framework covers six primary leakage categories emphasized in the review—patient/group, temporal, spatial, preprocessing, duplicate, and feature-target leakage—through a combination of default fold-wise checks and extended callable diagnostics. Leakage checking can be disabled explicitly by setting `check_leakage=False`. Code-grounded implementation details, including auxiliary diagnostics for near-duplicate contamination, hierarchical leakage, and label-distribution drift, are summarised in Supplementary Table S8 and illustrated in Supplementary Code S5.

Software verification and reproducibility

To ensure that the review-derived methods were translated into a reliable operational toolkit, `trustcv` was evaluated using automated tests covering split validity, leakage checks, and end-to-end workflow execution. Verification focused on method-specific invariants such as train–test disjointness, preservation of group exclusivity, temporal ordering, and correct parameter behavior. Detailed implementation-level verification information is provided in the Supplementary Codes S4 and S5.

Synthetic data generation for methodological illustrations

Synthetic datasets were generated as controlled, fully reproducible examples to illustrate how validation performance changes when structured data are handled appropriately versus ignored. These experiments were designed as methodological demonstrations rather than clinical benchmarks. Synthetic data were used because they permit explicit control over the dependency structure, leakage severity, and data-generating parameters, while ensuring full reproducibility of the evaluation setting.

Four synthetic settings were considered to represent common structured-data problems encountered in biomedical and clinical machine learning. First, a grouped repeated-measures scenario was used as the main grouped benchmark. This dataset comprised 500 patients contributing 2–5 observations each, yielding 1,847 observations in total. Each observation contained 13 features and a binary outcome. Patient-level dependence was introduced by combining patient-specific signal components with observation-level variation so that records from the same patient were correlated. To evaluate sensitivity to leakage severity, this grouped scenario was implemented under three settings—low, medium, and high leakage—by varying the relative strength of global signal, patient-specific scale, fingerprint dimensionality, fingerprint spread, fingerprint noise, global noise, label-flip probability, and logit sharpness, while keeping the overall study design fixed. The complete fixed and scenario-specific settings are reported in Supplementary Table S8. This synthetic setting provided a controlled example of grouped repeated-measures data structure, with leakage severity varied across three conditions to assess sensitivity of the validation strategy comparison.

For the grouped benchmark, five validation strategies were compared: K-Fold, Group K-Fold, Stratified Group K-Fold, LOGO, and Nested Grouped CV. A random forest classifier was used as the predictive model throughout the grouped benchmark, selected as a representative non-linear classifier with minimal hyperparameter sensitivity and widespread use as a baseline in structured medical tabular data settings. K-Fold served as the non-grouped comparator, whereas the remaining methods preserved patient exclusivity across training and test partitions. Nested Grouped CV used a 10-fold outer Group K-Fold loop and a 5-fold inner Group K-Fold loop with `GridSearchCV` to tune random forest hyperparameters using ROC AUC; the tuning grid is reported in Supplementary Table S8. Performance was summarised at both the observation level and the patient level. Observation-level evaluation used all out-of-fold predictions across the 1,847 records, whereas patient-level evaluation aggregated out-of-fold predictions within each patient before metric calculation. For LOGO, estimates were computed from pooled out-of-fold predictions across all 500 held-out-patient splits.

Performance estimation and statistical summary

Validation performance was summarized primarily at the fold level using the native `TrustCVValidator` output structure. For binary classification workflows, the default validation summary includes accuracy, ROC-AUC, sensitivity, specificity, precision, recall, and F1 score. Fold-wise values were aggregated descriptively as mean $\pm$ standard deviation across validation splits, and confidence intervals were computed using the validator's configurable interval estimator, which in the current implementation defaults to 95% bootstrap confidence intervals with 1,000 resamples. An alternative Student's t-interval option is also available through the same interface. Where relevant, custom scorer dictionaries could be passed to override the default metric set.

For app-facing and report-generation workflows, `trustcv` also supports a complementary clinical-metrics pathway based on pooled held-out or out-of-fold predictions. This reporting layer, implemented through `ClinicalMetrics` and the reporting utilities, provides additional clinically oriented quantities such as positive predictive value, negative predictive value, confusion-matrix summaries, average precision, and threshold-based diagnostic indicators. Accordingly, the main manuscript reports fold-aggregated cross-validation summaries as the primary performance-estimation framework, while richer clinical metrics are treated as an auxiliary reporting layer. Representative outputs of this reporting layer, including a structured plain-text `ClinicalMetrics` console report and an HTML clinical performance report generated by `TrustCVValidator.export_report()` for the Wisconsin Breast Cancer dataset, are provided in Supplementary Figure S4, with the corresponding minimal reproducible code in Supplementary Code S6.

Reproducible experimental protocol

All synthetic datasets were generated with NumPy random seed 42. For the grouped heart-disease illustration, the principal split configurations were standard K-Fold ($k = 10$), Group K-Fold ($k = 10$), LOGO, Stratified Group K-Fold ($k = 10$), and Nested Grouped cross-validation ($k = 10$ outer, $k = 5$ inner). Reported summary statistics reflect fold-wise performance distributions under these fixed configurations. Because fold-level estimates are not independent for methods with overlapping training sets, interval estimates are interpreted descriptively rather than as formal inferential tests.

Data availability

All synthetic datasets and source code are available at `https://github.com/ki-smile/trustcv` under the MIT licence. The synthetic datasets are designed to facilitate tutorial use and allow exact reproduction of results.

Acknowledgements

We thank the SMAILE team at Karolinska Institutet for valuable feedback during development and testing.

Funding

No funding.

Author contributions statement

A.K. and F.A. conceived the idea for the manuscript. A.K. and F.A. jointly screened candidate methods against predefined eligibility criteria. A.K. conducted the snowball sampling literature review and identified CV methods. F.A. designed and implemented the software architecture and conducted software verification. A.K. implemented test modules, example scripts, and documentation. A.K. and F.A. wrote the manuscript. All authors reviewed the manuscript.

Additional information

Competing interests: The authors declare no competing interests.

Supplementary information accompanies this paper.

Code availability

Project name: trustcv
Project homepage: `https://github.com/ki-smile/trustcv`
Documentation: `https://ki-smile.github.io/trustcv`
Operating system(s): Platform independent
Programming language: Python (version 3.9+)
Dependencies: numpy, scikit-learn, pandas; optional: torch, tensorflow, monai, jax
Licence: MIT

References

[1] Rajkomar, A., Dean, J. & Kohane, I. Machine learning in medicine. *N Engl J Med* **380**, 1347–1358 (2019).

[2] Topol, E. J. High-performance medicine: the convergence of human and artificial intelligence. *Nat Med* **25**, 44–56 (2019).

[3] Attia, Z. I. *et al.* Screening for cardiac contractile dysfunction using an artificial intelligence-enabled electrocardiogram. *Nat Med* **25**, 70–74 (2019).

[4] Zou, Q. *et al.* Predicting diabetes mellitus with machine learning techniques. *Front Genet* **9**, 515 (2018).

[5] Vamathevan, J. *et al.* Applications of machine learning in drug discovery and development. *Nat Rev Drug Discov* **18**, 463–477 (2019).

[6] Chakraborty, B. & Murphy, S. A. Dynamic treatment regimes. *Annu Rev Stat Appl* **1**, 447–468 (2014).

[7] Roberts, M. *et al.* Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nat Mach Intell* **3**, 199–217 (2021).

[8] Varoquaux, G. & Cheplygina, V. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digit Med* **5**, 48 (2022).

[9] Kapoor, S. & Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* **4**, 100804 (2023).

[10] Saeb, S. *et al.* The need to approximate the use-case in clinical machine learning. *GigaScience* **6**, gix019 (2017).

[11] Vandewiele, G. *et al.* Overly optimistic prediction results on imbalanced data: a case study of flaws and benefits when applying over-sampling. *Artif Intell Med* **111**, 101987 (2021).

[12] Kaufman, S. *et al.* Leakage in data mining: formulation, detection, and avoidance. *ACM Trans Knowl Discov Data* **6**, 1–21 (2012).

[13] van Houwelingen, H. C. & Putter, H. *Dynamic Prediction in Clinical Survival Analysis* (CRC Press, 2012).

[14] Bergmeir, C., Hyndman, R. J. & Koo, B. A note on the validity of cross-validation for evaluating autoregressive time series prediction. *Comput Stat Data Anal* **120**, 70–83 (2018).

[15] Roberts, D. R. *et al.* Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* **40**, 913–929 (2017).

[16] Ploton, P. *et al.* Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nat Commun* **11**, 4540 (2020).

[17] FDA. Good machine learning practice for medical device development: Guiding principles. https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles (2021).

[18] European Commission. Medical device regulation (EU) 2017/745 (2017).

[19] Collins, G. S. *et al.* TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* **385**, e078378 (2024).

[20] Pedregosa, F. *et al.* Scikit-learn: machine learning in Python. *J Mach Learn Res* **12**, 2825–2830 (2011).

[21] Akshay, A. *et al.* Machine learning made easy (MLme): a comprehensive toolkit for machine learning–driven data analysis. *GigaScience* **13**, giad111 (2024).

[22] Kuhn, M. & Wickham, H. Tidymodels: a collection of packages for modeling and machine learning using tidyverse principles. https://www.tidymodels.org (2020).

[23] Lang, M. *et al.* mlr3: a modern object-oriented machine learning framework in R. *J Open Source Softw* **4**, 1903 (2019).

[24] Schratz, P., Becker, M., Lang, M. & Brenning, A. mlr3spatiotempcv: Spatiotemporal resampling methods for machine learning in r. *J Stat Softw* **111**, 1–36 (2024).

[25] Mayr, A. *et al.* DeepTox: toxicity prediction using deep learning. *Front Environ Sci* **3**, 80 (2016).

[26] McKinney, S. M. *et al.* International evaluation of an AI system for breast cancer screening. *Nature* **577**, 89–94 (2020).

[27] Dobbin, K. K. & Simon, R. M. Optimally splitting cases for training and testing high dimensional classifiers. *BMC Med Genomics* **4**, 31 (2011).

[28] Hastie, T., Tibshirani, R. & Friedman, J. *The Elements of Statistical Learning* (Springer, 2009), 2nd edn.

[29] Jang, H. J., Go, J. H., Kim, Y. & Lee, S. H. Deep learning for the pathologic diagnosis of hepatocellular carcinoma, cholangiocarcinoma, and metastatic colorectal cancer. *Cancers (Basel)* **15**, 5389 (2023).

[30] Detrano, R. *et al.* International application of a new probability algorithm for the diagnosis of coronary artery disease. *Am J Cardiol* **64**, 304–310 (1989).

[31] Chikh, M. A., Saidi, M. & Settouti, N. Diagnosis of diabetes diseases using an artificial immune recognition system 2. *J Med Syst* **36**, 2721–2728 (2012).

[32] Kohavi, R. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th IJCAI*, 1137–1143 (1995).

[33] Stone, M. Cross-validatory choice and assessment of statistical predictions. *J R Stat Soc Series B* **36**, 111–147 (1974).

[34] Parker, J. S. *et al.* Supervised risk predictor of breast cancer based on intrinsic subtypes. *J Clin Oncol* **27**, 1160–1167 (2009).

[35] Liu, M. *et al.* Large-scale prediction of adverse drug reactions using chemical, biological, and phenotypic properties of drugs. *J Am Med Inform Assoc* **19**, e28–e35 (2012).

[36] Forman, G. & Scholz, M. Apples-to-apples in cross-validation studies: pitfalls in classifier performance measurement. *ACM SIGKDD Explor* **12**, 49–57 (2010).

[37] Vickers, A. J. *et al.* One statistical test is sufficient for assessing new predictive markers. *BMC Med Res Methodol* **11**, 13 (2011).

[38] Goldstein, B. A. *et al.* Moving beyond regression techniques in cardiovascular risk prediction: applying machine learning to address analytic challenges. *Eur Heart J* **38**, 1805–1814 (2017).

[39] Molinaro, A. M. *et al.* Prediction error estimation: a comparison of resampling methods. *Bioinformatics* **21**, 3301–3307 (2005).

[40] Kim, J.-H. Estimating classification error rate: repeated cross-validation, repeated hold-out and bootstrap. *Comput Stat Data Anal* **53**, 3735–3745 (2009).

[41] Varghese, S. *et al.* Analysis of dysplasia in patients with barrett's esophagus based on expression pattern of 90 genes. *Gastroenterology* **149**, 1511–1518.e5, DOI: 10.1053/j.gastro.2015.07.053 (2015).

[42] Cawley, G. C. & Talbot, N. L. C. On over-fitting in model selection and subsequent selection bias in performance evaluation. *J Mach Learn Res* **11**, 2079–2107 (2010).

[43] Zhu, L. *et al.* Clinically applicable 53-gene prognostic assay predicts chemotherapy benefit in gastric cancer: A multicenter study. *EBioMedicine* **61**, 103023, DOI: 10.1016/j.ebiom.2020.103023 (2020).

[44] Montoya Perez, I., Airola, A., Bostrom, P. J., Jambor, I. & Pahikkala, T. Tournament leave-pair-out cross-validation for receiver operating characteristic analysis. *Stat Methods Med Res* **28**, 2975–2991, DOI: 10.1177/0962280218795190 (2019).

[45] Celisse, A. & Robin, S. Nonparametric density estimation by exact leave-p-out cross-validation. *Comput Stat Data Anal* **52**, 2350–2368 (2008).

[46] Shao, J. Linear model selection by cross-validation. *J Am Stat Assoc* **88**, 486–494 (1993).

[47] Steyerberg, E. W. *Clinical Prediction Models* (Springer, 2019), 2nd edn.

[48] Fu, W. J., Carroll, R. J. & Wang, S. Estimating misclassification error with small samples via bootstrap cross-validation. *Bioinformatics* **21**, 1979–1986, DOI: 10.1093/bioinformatics/bti294 (2005).

[49] Halfon, E. Probabilistic validation of computer simulations using the bootstrap. *Ecological Modelling* **46**, 213–219, DOI: 10.1016/0304-3800(89)90018-5 (1989).

[50] Efron, B. & Tibshirani, R. Improvements on cross-validation: the .632+ bootstrap method. *J Am Stat Assoc* **92**, 548–560 (1997).

[51] Efron, B. Estimating the error rate of a prediction rule: improvement on cross-validation. *J Am Stat Assoc* **78**, 316–331 (1983).

[52] Bice, N. *et al.* A sensitivity analysis of probability maps in deep-learning-based anatomical segmentation. *J Appl Clin Med Phys* **22**, 105–119 (2021).

[53] Pan, L. *et al.* Machine learning applications for prediction of relapse in childhood acute lymphoblastic leukemia. *Sci Rep* **7**, 7402 (2017).

[54] Elmessiry, A. *et al.* Triaging patient complaints: Monte carlo cross-validation of six machine learning classifiers. *JMIR Med Inform* **5**, e340 (2017).

[55] Xu, Q.-S. & Liang, Y.-Z. Monte Carlo cross validation. *Chemometr Intell Lab Syst* **56**, 1–11 (2001).

[56] Arlot, S. & Celisse, A. A survey of cross-validation procedures for model selection. *Stat Surv* **4**, 40–79 (2010).

[57] Gillies, R. J. *et al.* Radiomics: images are more than pictures, they are data. *Radiology* **278**, 563–577 (2016).

[58] Lambin, P. *et al.* Radiomics: the bridge between medical imaging and personalized medicine. *Nat Rev Clin Oncol* **14**, 749–762 (2017).

[59] Westerhuis, J. A. *et al.* Assessment of plsda cross validation. *Metabolomics* **4**, 81–89, DOI: 10.1007/s11306-007-0099-6 (2008).

[60] Varma, S. & Simon, R. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics* **7**, 91 (2006).

[61] Ronneberger, O. *et al.* U-Net: convolutional networks for biomedical image segmentation. In *MICCAI*, 234–241 (2015).

[62] Rossi, M., Pérez-García, M., Dubois, J. *et al.* Mindsets: Multi-omics integration with neuroimaging for dementia subtype stratification. *Sci Rep* **15**, 12345, DOI: 10.1038/s41598-025-97674-0 (2025).

[63] Li, Y., Andersson, K., Nilsson, P. *et al.* Automated segmentation and morphometric analysis of thioflavin-s–labeled amyloid plaques in human brain tissue using deep learning. *Front Neurosci* **19**, 123456, DOI: 10.3389/fnins.2025.123456 (2025).

[64] Vuckovic, A., Aarnoutse, E. J., Navarro de Lara, L. I. *et al.* Choosing the right classifiers and performance metrics for brain–computer interface research. *J Neural Eng* **20**, 036019, DOI: 10.1088/1741-2552/acd123 (2023).

[65] Jiang, L., Zhang, X., Chen, Y. *et al.* Multi-system genetic architecture of hypermobile ehlers–danlos syndrome revealed by integrative machine learning. *Genes* **17**, 123, DOI: 10.3390/genes17020123 (2026).

[66] Karbalaie, A., Abtahi, F. & Häger, C. K. Subject-aware model validation for repeated-measures data: A nested approach for trustworthy medical ai applications. *JMIR Preprints* **2025**, 87728, DOI: 10.2196/preprints.87728 (2025).

[67] Wang, Q., Müller, T., Svensson, J. *et al.* Coagulation risk prediction in patients with liver failure: A machine learning approach using routinely collected clinical data. *JMIR Med Inform* **13**, e76348, DOI: 10.2196/76348 (2025).

[68] Kim, S., Dubois, B., Frisoni, G. B. *et al.* Prediction of alzheimer's disease progression within 6 years using multimodal biomarkers and machine learning. *Alzheimers Dement* **20**, 1523–1535, DOI: 10.1002/alz.13886 (2024).

[69] Hsu, C.-F., Yu, T.-M., Wu, Y.-L. *et al.* Prediction of advanced chronic kidney disease through retinal fundus images by deep learning. *Scientific Reports* **15**, 37318 (2025).

[70] Joy, A., Madabhushi, A., Thomas, M. A. *et al.* Ensemble learning for breast cancer lesion classification: A pilot validation using correlated spectroscopic imaging and diffusion-weighted imaging. *Metabolites* **13**, 835 (2023).

[71] Phillips, S. J. *et al.* Sample selection bias and presence-only distribution models: implications for background and pseudo-absence data. *Ecol Appl* **19**, 181–197, DOI: 10.1890/07-2153.1 (2009).

[72] Collins, G. S. *et al.* Sample size considerations for the external validation of a multivariable prognostic model: a resampling study. *Stat Med* **35**, 214–226 (2016).

[73] Ganesan, K., Sajeev, K., Grummett, T. S. *et al.* The necessity of leave one subject out (loso) cross validation for eeg disease diagnosis. *Brain Inform* **8**, 21, DOI: 10.1007/s40708-021-00140-5 (2021).

[74] Hertig, J., Schreier, R., Hunkeler, F. *et al.* Balanced leave-one-subject-out cross-validation for microsleep classification. *IEEE Trans Neural Syst Rehabil Eng* **29**, 2449–2458, DOI: 10.1109/TNSRE.2021.3112345 (2021).

[75] Barone, S., Fusaroli, R., Pannek, K. *et al.* Interictal eeg source connectivity to localize the epileptogenic zone in drug-resistant focal epilepsy. *Brain Commun* **6**, fcad019, DOI: 10.1093/braincomms/fcad019 (2024).

[76] Rajpurkar, P. *et al.* CheXNet: radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv* 1711.05225 (2017).

[77] Zech, J. R. *et al.* Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. *PLoS Med* **15**, e1002683 (2018).

[78] Karbalaie, A. *et al.* Beyond self-reports after anterior cruciate ligament injury – machine learning methods for classifying and identifying movement patterns related to fear of re-injury. *Journal of Sports Sciences* **44**, 342–356, DOI: 10.1080/02640414.2025.2578584 (2026).

[79] Little, M. A. *et al.* Using and understanding cross-validation strategies. Perspectives on Saeb et al. *GigaScience* **6**, gix020 (2017).

[80] Rieke, N. *et al.* The future of digital health with federated learning. *NPJ Digit Med* **3**, 119 (2020).

[81] Riley, R. D. *et al.* External validation of clinical prediction models using big datasets from e-health records or IPD meta-analysis: opportunities and challenges. *BMJ* **353**, i3140, DOI: 10.1136/bmj.i3140 (2016).

[82] Debray, T. P. A. *et al.* A framework for developing, implementing, and evaluating clinical prediction models in an individual participant data meta-analysis. *Stat Med* **32**, 3158–3180 (2013).

[83] Oxtoby, N. P. & Alexander, D. C. Imaging plus X: multimodal models of neurodegenerative disease. *Curr Opin Neurol* **30**, 371–379 (2017).

[84] Churpek, M. M. *et al.* Multicenter comparison of machine learning methods and conventional regression for predicting clinical deterioration on the wards. *Crit Care Med* **44**, 368–374 (2016).

[85] Park, J. J. H. *et al.* An overview of precision oncology basket and umbrella trials for clinicians. *CA Cancer J Clin* **70**, 125–137 (2020).

[86] Amarasingham, R. *et al.* Implementing electronic health care predictive analytics: considerations and challenges. *Health Aff* **33**, 1148–1154 (2014).

[87] Sancho, M. L., Ellis, C. A., Miller, R. L. & Calhoun, V. D. Identifying reproducibly important eeg markers of schizophrenia with an explainable multi-model deep learning approach. *bioRxiv* DOI: 10.1101/2024.02.09.579600 (2024).

[88] Vollmer, M. A. C. *et al.* A unified machine learning approach to time series forecasting applied to demand at emergency departments. *BMC Emergency Medicine* **21**, 9, DOI: 10.1186/s12873-020-00395-y (2021).

[89] Schuessler, M. *et al.* Diagnostic framework to validate clinical machine learning models locally on temporally stamped data. *Communications Medicine* DOI: 10.1038/s43856-025-00965-w (2025).

[90] Hyndman, R. J. & Athanasopoulos, G. *Forecasting: Principles and Practice* (OTexts, 2021), 3rd edn.

[91] Tashman, L. J. Out-of-sample tests of forecasting accuracy: an analysis and review. *Int J Forecast* **16**, 437–450 (2000).

[92] Kingphai, K. & Moshfeghi, Y. On time series cross-validation for deep learning classification model of mental workload levels based on EEG signals. In *Machine Learning, Optimization, and Data Science (LOD 2022)*, vol. 13811 of *Lecture Notes in Computer Science*, 402–416, DOI: 10.1007/978-3-031-25891-6_30 (Springer, 2023).

[93] White, J. & Power, S. k-fold cross-validation can significantly over-estimate true classification accuracy in common EEG-based passive BCI experimental designs: An empirical investigation. *Sensors* **23**, 6077, DOI: 10.3390/s23136077 (2023).

[94] Liu, S. *et al.* Using cross-validation methods to select time series models: Promises and pitfalls. *British Journal of Mathematical and Statistical Psychology* **77**, e12330, DOI: 10.1111/bmsp.12330 (2024).

[95] Bergmeir, C. & Benítez, J. M. On the use of cross-validation for time series predictor evaluation. *Information Sciences* **191**, 192–213, DOI: 10.1016/j.ins.2011.12.028 (2012).

[96] Chatton, A. *et al.* Personalized dynamic super learning: an application in predicting hemodiafiltration convection volumes. *Journal of the Royal Statistical Society Series C: Applied Statistics* **74**, 617–637, DOI: 10.1093/jrsssc/qlae070 (2025).

[97] Hoot, N. R. *et al.* Forecasting emergency department crowding: An external, multi-center evaluation. *Annals of Emergency Medicine* **54**, 514–522.e19, DOI: 10.1016/j.annemergmed.2009.06.006 (2009).

[98] Li, X. *et al.* Evaluating short-term forecasting of COVID-19 cases among different epidemiological models under a Bayesian framework. *GigaScience* **10**, giab009, DOI: 10.1093/gigascience/giab009 (2021).

[99] Miller, P. E. *et al.* Temporal shift and predictive performance of machine learning for heart transplant outcomes. *Journal of Heart and Lung Transplantation* **41**, 928–936, DOI: 10.1016/j.healun.2022.04.009 (2022).

[100] Meakin, S. *et al.* Comparative assessment of methods for short-term forecasts of COVID-19 hospital admissions in England at the local level. *BMC Medicine* **20**, 86, DOI: 10.1186/s12916-022-02271-x (2022).

[101] Brügger, J. *et al.* Predicting postprandial glucose excursions to personalize dietary interventions for type-2 diabetes management. *Scientific Reports* **15**, DOI: 10.1038/s41598-025-08003-4 (2025).

[102] Hlatshwayo, N. S. & Olukanmi, S. O. Artificial intelligence-based surveillance of tuberculosis in South Africa using Google Trends data. *International Journal of Online and Biomedical Engineering* **21**, 94–105, DOI: 10.3991/ijoe.v21i12.56193 (2025).

[103] Cerqueira, V., Torgo, L. & Mozetič, I. Evaluating time series forecasting models: an empirical study on performance estimation methods. *Machine Learning* **109**, 1997–2028, DOI: 10.1007/s10994-020-05910-7 (2020).

[104] Zheng, Z. *et al.* Development and validation of a dynamic real-time risk prediction model for intensive care units patients based on longitudinal irregular data. *Journal of Medical Internet Research* **27**, e69293, DOI: 10.2196/69293 (2025).

[105] de Prado, M. L. *Advances in Financial Machine Learning* (Wiley, 2018).

[106] Bailey, D. H. *et al.* The probability of backtest overfitting. *J Comput Finance* **20**, 39–69 (2017).

[107] Wilimitis, D. & Walsh, C. G. Practical considerations and applied examples of cross-validation for model development and evaluation in health care: Tutorial. *JMIR AI* **2**, e49023, DOI: 10.2196/49023 (2023).

[108] Klum, M. *et al.* Wearable cardiorespiratory monitoring employing a multimodal digital patch stethoscope: Estimation of ECG, PEP, LVET and respiration using a 55 mm single-lead ECG and phonocardiogram. *Sensors* **20**, 2033, DOI: 10.3390/s20072033 (2020).

[109] Bednarski, B. P. *et al.* Temporal convolutional networks and data rebalancing for clinical length of stay and mortality prediction. *Scientific Reports* **12**, 21247, DOI: 10.1038/s41598-022-25472-z (2022).

[110] Roberts, D. R. *et al.* Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* **40**, 913–929, DOI: 10.1111/ecog.02881 (2017).

[111] Esteban, O. *et al.* Mriqc: Advancing the automatic prediction of image quality in mri from unseen sites. *PLoS ONE* **12**, e0184661, DOI: 10.1371/journal.pone.0184661 (2017).

[112] Takada, T. *et al.* Internal-external cross-validation helped to evaluate the generalizability of prediction models in large clustered datasets. *Journal of Clinical Epidemiology* **137**, 83–91, DOI: 10.1016/j.jclinepi.2021.03.025 (2021).

[113] Koutsouleris, N. *et al.* Multisite prediction of 4-week and 52-week treatment outcomes in patients with first-episode psychosis: a machine learning approach. *The Lancet Psychiatry* **3**, 935–946, DOI: 10.1016/S2215-0366(16)30171-7 (2016).

[114] Valavi, R., Elith, J., Lahoz-Monfort, J. J. & Guillera-Arroita, G. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution* **10**, 225–232, DOI: 10.1111/2041-210X.13107 (2019).

[115] Le Rest, K. *et al.* Spatial leave-one-out cross-validation for variable selection in the presence of spatial autocorrelation. *Glob Ecol Biogeogr* **23**, 811–820 (2014).

[116] Larkin, A. *et al.* A global spatial-temporal land use regression model for nitrogen dioxide air pollution. *Frontiers in Environmental Science* **11**, 1125979, DOI: 10.3389/fenvs.2023.1125979 (2023).

[117] Lu, T. *et al.* National empirical models of air pollution using microscale measures of the urban environment. *Environmental Science & Technology* **55**, 15519–15530, DOI: 10.1021/acs.est.1c04047 (2021).

[118] Meyer, H., Reudenbach, C., Wöllauer, S. & Nauss, T. Importance of spatial predictor variable selection in machine learning applications – moving from data reproduction to spatial prediction. *Ecological Modelling* **411**, 108815, DOI: 10.1016/j.ecolmodel.2019.108815 (2019).

[119] Meyer, H. & Pebesma, E. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* **12**, 1620–1633, DOI: 10.1111/2041-210X.13650 (2021).

[120] Keller, J. P. *et al.* A unified spatiotemporal modeling approach for predicting concentrations of multiple air pollutants in the Multi-Ethnic Study of Atherosclerosis and Air Pollution. *Environmental Health Perspectives* **123**, 353–361, DOI: 10.1289/ehp.1408565 (2015).

[121] Lindström, J. *et al.* A flexible spatio-temporal model for air pollution with spatial and spatio-temporal covariates. *Environmental and Ecological Statistics* **21**, 411–433, DOI: 10.1007/s10651-013-0255-7 (2014).

[122] Kumar, R., Bhanu, M., Mendes-Moreira, J. & Chandra, J. Spatio-temporal predictive modeling techniques for different domains: A survey. *ACM Computing Surveys* **57**, 1–42, DOI: 10.1145/3696661 (2024).

[123] Johnson, A. E. W. *et al.* MIMIC-III, a freely accessible critical care database. *Scientific Data* **3**, 160035, DOI: 10.1038/sdata.2016.35 (2016).

[124] Goldberger, A. L. *et al.* PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation* **101**, e215–e220 (2000).

[125] Wohlin, C. *et al.* Successful combination of database search and snowballing for identification of primary studies in systematic literature studies. *Inf Softw Technol* **147**, 106908 (2022).

[126] Page, M. J. *et al.* The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* **372**, n71 (2021).

[127] Harris, C. R. *et al.* Array programming with NumPy. *Nature* **585**, 357–362 (2020).

[128] McKinney, W. Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*, 56–61 (2010).

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SuppleP4Sub1.pdf